

Unit 3(Part 2)

Quality of Service (QoS) concepts and protocols, and network security fundamentals

Introduction to QoS

Quality of Service (QoS) refers to the **ability of a network to provide better service to selected network traffic** over various technologies like IP, ATM, Frame Relay, etc.

It ensures that **important data gets priority**, especially in real-time applications like:

- Voice over IP (VoIP)
- Video conferencing
- Online gaming

Why QoS is Needed?

Traditional networks follow **best-effort delivery**, meaning:

- No guarantee of delivery time
- No guarantee of bandwidth
- No guarantee of packet order

QoS is required to:

- Avoid congestion
- Ensure timely delivery
- Reduce packet loss
- Provide guaranteed performance

QoS PERFORMANCE PARAMETERS

1. Bandwidth

- Maximum data transfer capacity of a network

- Example:
If a link has **10 Mbps bandwidth**, it can carry 10 million bits/sec

2. Delay (Latency)

Total time taken by a packet:

Total Delay = Processing + Queuing + Transmission + Propagation

Breakdown:

- **Processing Delay** → Time to check packet header
- **Queuing Delay** → Waiting time in router queue
- **Transmission Delay** → Time to push bits onto link
- **Propagation Delay** → Time to travel through medium

Example:

A packet waits 2 ms in queue + 1 ms transmission → total delay = 3 ms

3. Jitter

- Variation in delay between packets
Example:
Packet 1 delay = 10 ms
Packet 2 delay = 30 ms
Jitter = 20 ms (bad for video/audio)

4. Packet Loss

- Packets dropped due to congestion
Example:
100 packets sent, 5 lost → loss = 5%

5. Throughput

- Actual successful delivery rate
Example:
Bandwidth = 10 Mbps but throughput = 7 Mbps due to congestion

QoS SCHEDULING ALGORITHMS

Scheduling decides: “Which packet should be transmitted next?”

Used in routers to:

- Reduce delay
- Handle priority traffic
- Ensure fairness

1. FIFO (First In First Out)

Concept

- Packets are served in the order they arrive
- Also called FCFS (First Come First Serve)

Working

1. Packets enter queue
2. Stored sequentially
3. First packet is transmitted first

Example

Arrival order:

- P1 (Video)
- P2 (Email)
- P3 (Voice)

Output: P1 → P2 → P3 (No priority considered)

Advantages

- Simple
- Low overhead
- Easy to implement

Disadvantages

- No priority handling
- High delay for important packets
- Not suitable for QoS

2. Priority Queuing (PQ)

- Packets are classified into different priority levels
- Higher priority packets are served first

Working

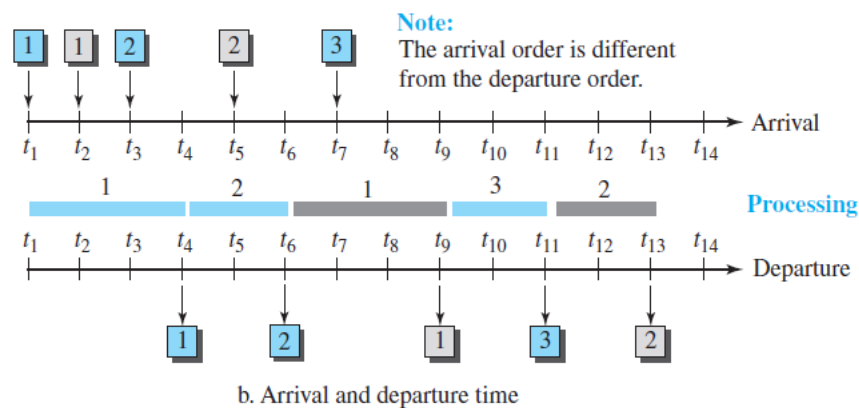
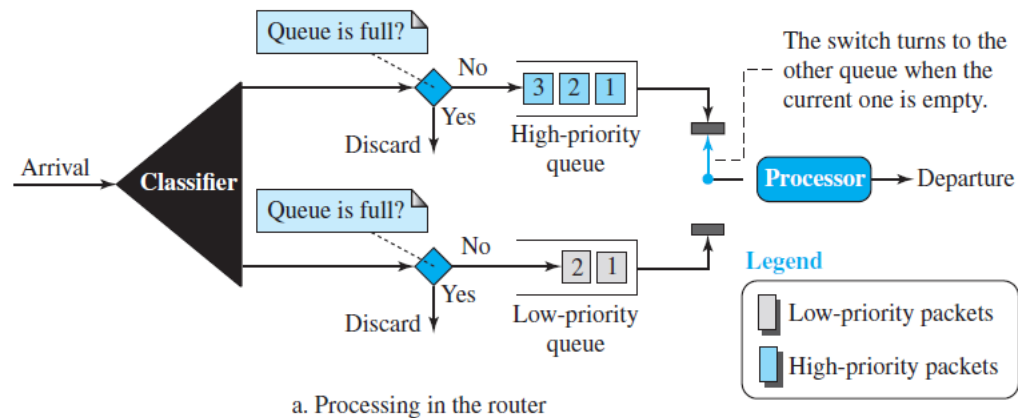
1. Multiple queues created:
 - High priority
 - Medium
 - Low
2. Scheduler always serves:
High → Medium → Low

Example

Packet	Type	Priority
P1	Voice	High
P2	Email	Low
P3	Video	Medium

Output: P1 → P3 → P2

Figure 30.2 Priority queuing



Advantages

- Important traffic gets priority
- Good for real-time apps

Disadvantages

- Starvation problem(Low priority packets may never be served)

3. Weighted Fair Queuing (WFQ)

- Provides fair bandwidth allocation
- Based on weights assigned to flows

Working

- Each flow gets share proportional to its weight.

- In this technique, the packets are still assigned to different classes and admitted to different queues.
- The queues, however, are weighted based on the priority of the queues; higher priority means a higher weight.
- The system processes packets in each queue in a round-robin fashion with the number of packets selected from each queue based on the corresponding weight.
- For example, if the weights are 3, 2, and 1, three packets are processed from the first queue, two from the second queue, and one from the third queue. In this way, we have fair queuing with priority

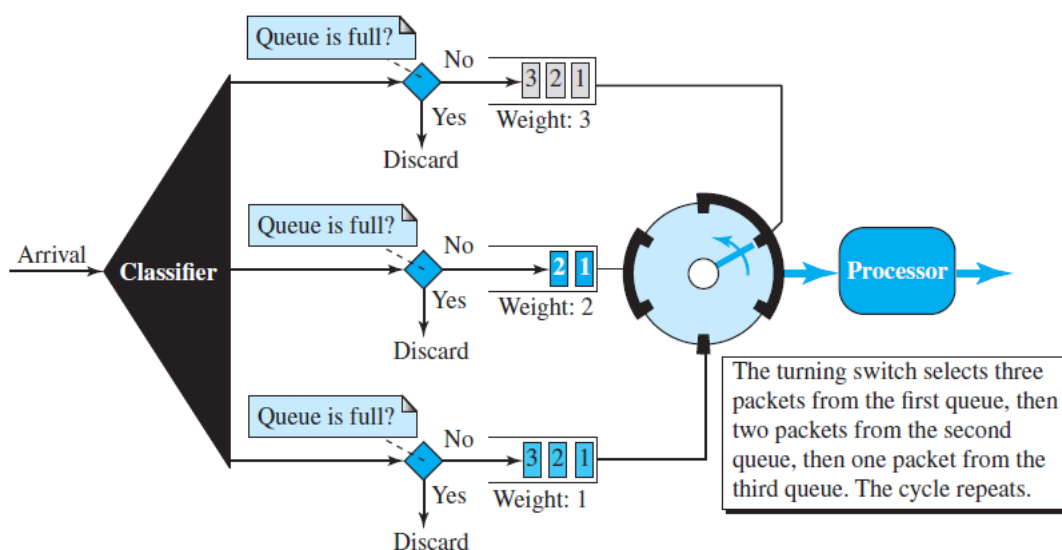
Example

Flow	Weight
A	2
B	1

Bandwidth division:

- A gets $2/3$
- B gets $1/3$

Figure 30.3 Weighted fair queuing



Advantages

- Fair distribution
- Prevents starvation
- Supports QoS

Disadvantages

- Complex implementation

TRAFFIC SHAPING or POLICING(Important)

To control the amount and the rate of traffic is called *traffic shaping* or *traffic policing*.

The first term is used when the traffic leaves a network; the second term is used when the data enters the network. Two techniques can shape or police the traffic: leaky bucket and token bucket.

1. LEAKY BUCKET ALGORITHM

Leaky Bucket is a **traffic shaping algorithm** used in QoS to:

- Control data transmission rate
- Reduce congestion
- Smooth bursty traffic

Thus, “Data enters the bucket at any rate but leaves at a **fixed constant rate**.”

Real-Life Analogy

Imagine a **bucket with a small hole at the bottom**:

- Water enters randomly (fast/slow)
- Water leaks out at a constant rate
- If bucket overflows → water is lost

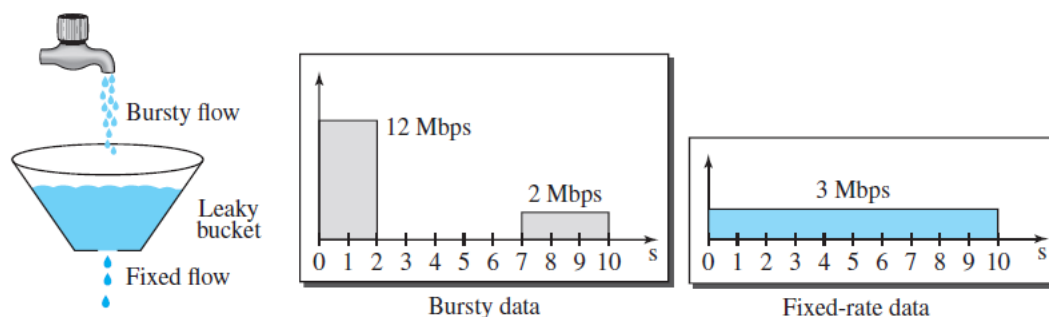
Same happens with packets:

- Incoming traffic = water entering
- Outgoing traffic = leak
- Overflow = packet drop

3. Working of Leaky Bucket Algorithm

Components

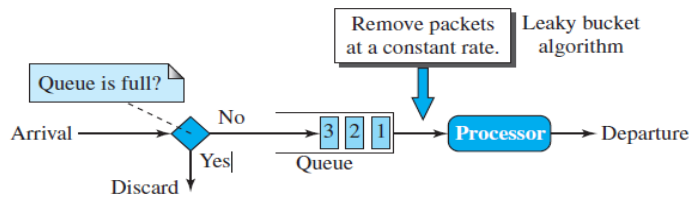
- Bucket (Buffer/Queue)
- Output link (fixed rate)
- Incoming packets



In the figure, we assume that the network has committed a bandwidth of 3 Mbps for a host. The use of the leaky bucket shapes the input traffic to make it conform to this commitment. In Figure 30.4 the host sends a burst of data at a rate of 12 Mbps for 2 seconds, for a total of 24 Mb of data. The host is silent for 5 seconds and then sends data at a rate of 2 Mbps for 3 seconds, for a total of 6 Mb of data. In all, the host has sent 30 Mb of data in 10 seconds. The leaky bucket smooths the traffic by sending out data at a rate of 3 Mbps during the same 10 seconds. Without the leaky bucket, the beginning burst may have hurt the network by consuming more bandwidth than is set aside for this host. We can also see that the leaky bucket may prevent congestion.

Step-by-Step Working

1. Packets arrive at variable rate
2. Stored in bucket (queue)
3. Packets are transmitted at constant rate
4. If bucket is full → extra packets are dropped



The following is an algorithm for variable-length packets:

1. Initialize a counter to n at the tick of the clock.
2. If n is greater than the size of the packet, send the packet and decrement the counter by the packet size. Repeat this step until the counter value is smaller than the packet size.
3. Reset the counter to n and go to step 1.

A leaky bucket algorithm shapes bursty traffic into fixed-rate traffic by averaging the data rate. It may drop the packets if the bucket is full.

Let:

- **B = Bucket capacity (in packets or bytes)**
- **R = Output rate (packets/sec)**
- **Input rate = variable**

Condition:

- If incoming data $> B \rightarrow$ overflow $\rightarrow$ packet loss

Example

Given:

- Bucket size = 5 packets
- Output rate = 1 packet/sec

Working

- At time $t = 0$, 8 packets arrive
- Bucket can hold only 5 packets
- Remaining 3 packets will be dropped

Transmission:

- $t=1 \rightarrow$ 1 packet sent

- $t=2 \rightarrow 1$ packet sent
- $t=3 \rightarrow 1$ packet sent
- $t=4 \rightarrow 1$ packet sent
- $t=5 \rightarrow 1$ packet sent

Thus, Output becomes smooth and constant

Advantages

- Smooth and constant traffic flow
- Easy to implement
- Prevents sudden congestion
- Suitable for networks needing steady rate

Disadvantages

- **Packet loss occurs** when bucket is full
- Does not allow burst transmission
- Not suitable for real-time bursty applications
- Rigid (no flexibility)

2. TOKEN BUCKET

Token Bucket is a **traffic shaping algorithm** used in QoS that:

- Controls data transmission rate
- Allows **burst traffic**
- Prevents congestion

Main Idea: "Packets can be sent only if there are sufficient tokens available."

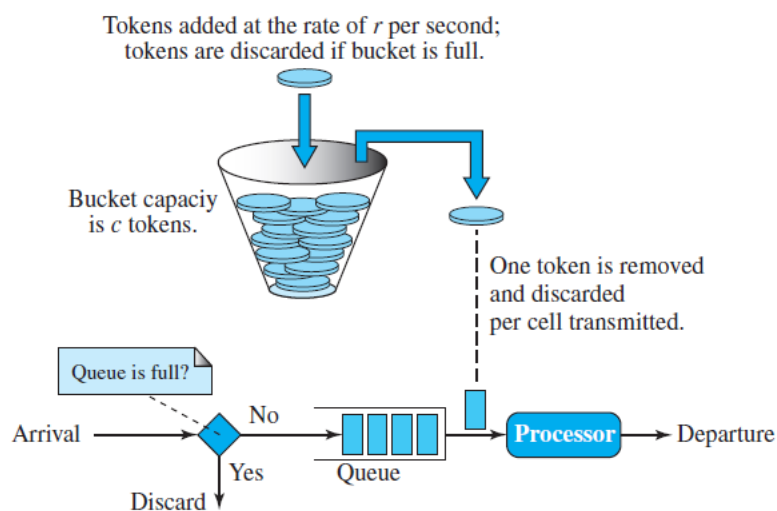
Key Components

- **Bucket** $\rightarrow$ stores tokens (not packets)
- **Tokens** $\rightarrow$ permission to send data
- **Token generation rate (R)** $\rightarrow$ tokens/sec
- **Bucket capacity (B)** $\rightarrow$ maximum tokens stored

Working of Token Bucket

- Tokens are generated at a **fixed rate (R)**
- Tokens are stored in the bucket (max capacity B)
- When a packet arrives:
 - If tokens available → packet is sent
 - If no tokens → packet waits (or dropped in policing)
- Unused tokens accumulate → allows burst

Figure 30.6 *Token bucket*



Mathematical Representation

Let:

- **R = Token generation rate (tokens/sec)**
- **B = Bucket capacity (tokens)**
- **Packet size = tokens required**

Condition: Packet can be transmitted only if enough tokens exist

Example

Given:

- Token rate = 2 tokens/sec
- Bucket size = 5 tokens

Case 1: Idle Network

- No packets for 3 sec
- Tokens accumulated = $2 \times 3 = 6$
- But max capacity = 5 $\rightarrow$ bucket full

Now burst of 5 packets can be sent instantly

Case 2: Continuous Traffic

- Packets arrive continuously
- Only 2 packets/sec can be sent

Burst Handling

Maximum burst size = **Bucket capacity (B)**

After burst: **Transmission continues at rate R**

Time	Tokens Available	Action
t=0	5	Burst allowed
t=1	2 added	Send packets
t=2	2 added	Continue

Advantages

- Allows burst transmission
- Efficient bandwidth utilization
- Less packet loss
- Flexible and realistic

Disadvantages

- Slightly complex
- Requires token management

Where It Is Used

- Routers and switches
- Internet traffic control

- Video streaming systems
- QoS mechanisms

NUMERICALS ON TRAFFIC SHAPING ALGORITHMS

Question 1:

A network uses the Token Bucket algorithm with the following parameters:

- Token generation rate = 4 tokens/sec
- Bucket capacity = 8 tokens
- Initially, the bucket is full

Answer the following:

1. How many packets can be sent immediately if 10 packets arrive at time $t = 0$?
2. How many packets will be sent at time $t = 1$ sec if 6 new packets arrive?
3. What is the maximum burst size?

Solution:

Initial Condition

- Bucket capacity = 8 tokens
- Initially full $\rightarrow$ Tokens available = 8

1. At $t = 0$

- Packets arrived = 10
- Tokens available = 8

Only 8 packets can be sent, Remaining 2 packets will wait

2. At $t = 1$ sec

- Tokens generated = 4 tokens
- Previous leftover packets = 2
- New packets = 6

Total packets = $2 + 6 = 8$

Tokens available = 4

Only 4 packets can be sent, Remaining packets = 4

3. Maximum Burst Size

Maximum burst = Bucket capacity = 8 packets

Final Answer:

1. 8 packets sent immediately
2. 4 packets sent at $t = 1$ sec
3. Maximum burst size = 8 packets

Question 2:

A system uses the Leaky Bucket algorithm with:

- Bucket size = 5 packets
- Output rate = 1 packet/sec

At time $t = 0$, 9 packets arrive.

Answer the following:

1. How many packets are dropped?
2. How long will it take to transmit all packets?
3. Draw the packet transmission timeline.

Solution:

- Bucket size = 5 packets
- Incoming packets = 9

Only 5 packets stored,

Therefore, Dropped packets = $9 - 5 = 4$

Output rate = 1 packet/sec

5 packets $\times$ 1 sec = **5 seconds**

Question3:

Compare the behaviour of Leaky Bucket and Token Bucket for the following:

- Incoming burst = 6 packets at $t = 0$
- Leaky Bucket:
 - Capacity = 4 packets
 - Output rate = 1 packet/sec
- Token Bucket:
 - Token rate = 2 tokens/sec
 - Bucket size = 4 tokens

Find:

1. Packets transmitted immediately
2. Packet loss (if any)
3. Which algorithm performs better?

Solution:

Leaky Bucket

- Capacity = 4
- Incoming = 6

Stored = 4, Dropped = 2

Output: 1 packet/sec, Total time = 4 sec

Token Bucket

- Bucket size = 4 tokens
- Initially full

Immediate transmission = 4 packets

Remaining = 2 packets (wait)

At $t = 1$ sec:

- Tokens generated = 2, Remaining 2 packets sent

Therefore,

1. Both send 4 packets immediately
2. Leaky Bucket drops 2 packets; Token Bucket drops none

3. Token Bucket performs better

SERVICES In QoS

1. Integrated Services (IntServ)

- Provides **end-to-end QoS guarantee for each individual flow**
- Works on **per-flow basis**

Idea: “Reserve resources for every flow before transmission.”

Key Features

- Per-flow resource reservation
- Guaranteed bandwidth and delay
- Uses signaling protocol

Protocol Used: RSVP (Resource Reservation Protocol)

Working

1. Sender sends **PATH message(RSVP Message)**
2. Receiver replies with **RESV message(RSVP Message)**
3. Routers reserve resources (bandwidth, buffers)
4. Data transmission starts with guaranteed QoS

RSVP uses Path messages. A Path message travels from the sender and reaches all receivers in the multicast path. On the way, a Path message stores the necessary information for the receivers. A Path message is sent in a multicast environment; a new message is created when the path diverges

After a receiver has received a Path message, it sends a Resv message. The Resv message travels toward the sender (upstream) and makes a resource reservation on the routers that support RSVP.

Thus,

- **PATH message: sent by sender to receiver to establish path and send traffic information**
- **RESV message: sent by receiver to sender to reserve resources along the path**

Figure 30.7 *Path messages*

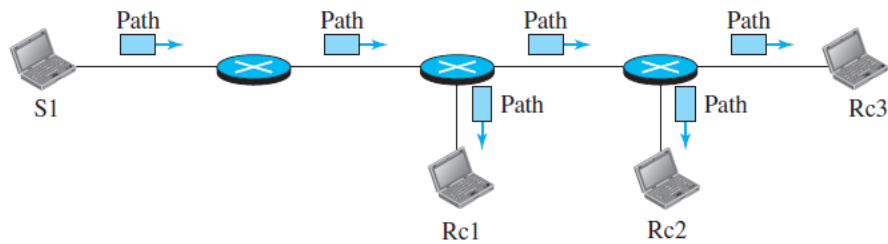


Figure 30.8 *Resv messages*

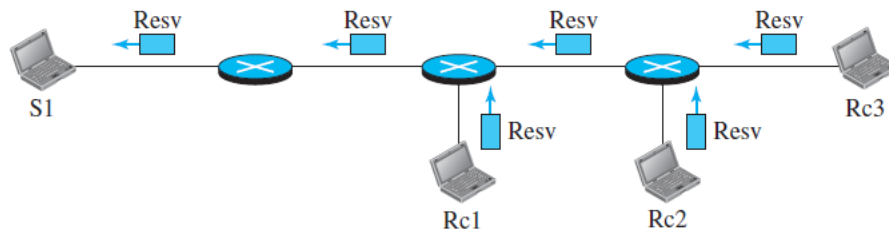
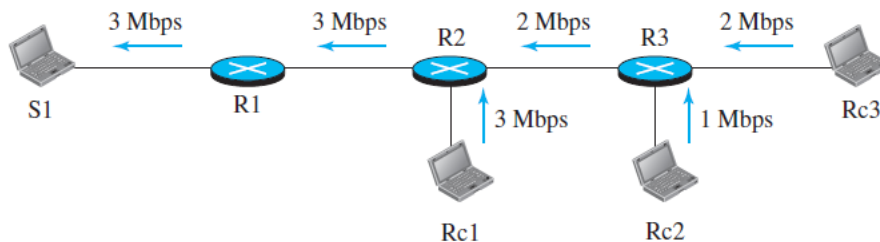


Figure 30.9 *Reservation merging*



Advantages

- Strong QoS guarantee
- Suitable for real-time applications
- Ensures low delay and no packet loss

Disadvantages

- **Not scalable** (too many flows)
- High overhead
- Complex implementation

Example: Video call:

- Bandwidth reserved before communication

- Smooth, uninterrupted transmission

2. Differentiated Services (DiffServ)

- Provides QoS based on **traffic classes (not individual flows)**
- Works on **per-class basis**

Idea: “Treat packets differently based on priority class.”

Key Features

- No per-flow reservation
- Packets classified into classes
- Uses DS field in IP header

What is DS Field?

A special field in the **IP header** used to:

- Identify packet priority
- Define QoS treatment

Structure of DS Field (8 bits)

| 6 bits (DSCP) | 2 bits (Unused) |

DSCP = Differentiated Services Code Point

These 6 bits decide:

- Priority level
- Type of service

Working

1. Packets are classified (e.g., voice, video, data)
2. Marked using **DS field**
3. Routers apply different treatment:
 - High priority → fast forwarding
 - Low priority → normal service

PER-HOP BEHAVIOR (PHB)

PHB defines: “How a router treats a packet based on DSCP value”

Types of PHB

1. Expedited Forwarding (EF)

- Highest priority
- Low delay, low loss

Used for: Voice calls , Real-time video

2. Assured Forwarding (AF)

- Medium priority
- Provides delivery assurance

Used for: Streaming , Business applications

3. Best Effort

- Lowest priority
- No guarantee

Used for: Email , Normal browsing

Advantages

- Highly scalable
- Simple implementation
- Efficient for large networks

Disadvantages

- No strict QoS guarantee
- Only relative priority (not absolute guarantee)

Example

Internet traffic:

- Video streaming → high priority
- Email → low priority

Feature	IntServ	DiffServ
Type	Per-flow	Per-class
QoS Guarantee	Strong	Weak (relative)
Scalability	Low	High
Complexity	High	Low
Protocol	RSVP	No reservation protocol
Resource Reservation	Yes	No
Overhead	High	Low
Implementation	Complex	Simple

Network Security Fundamentals

SECURITY GOALS

Network security aims to protect data and communication using multiple goals:

1.1 Confidentiality

- Ensures that **information is not disclosed to unauthorized users**
- Protects sensitive data like passwords, bank details, personal information

Techniques:

- **Encryption**
 - Plaintext → Ciphertext
- **Access Control**
 - Only authorized users can access data

Example:

When you use online banking, your data is encrypted so attackers cannot read it.

1.2 Integrity

- Ensures that **data remains accurate and unaltered**
- Protects against unauthorized modification

Techniques:

- Hash Functions (SHA, MD5)
- Digital Signatures

Example:

Message sent: “*Transfer ₹500*”

If changed to “*₹5000*”, integrity is violated

1.3 Availability

- Ensures that **network services are available when required**

Threats:

- DoS (Denial of Service)
- Hardware failure

Techniques:

- Backup systems
- Redundancy
- Load balancing

1.4 Authentication

- Verifying **identity of user or system**

Types:

1. Something you know → Password
2. Something you have → ATM card
3. Something you are → Biometrics

Example:

Login using OTP + password

1.5 Authorization (Access Control)

- Determines **what an authenticated user is allowed to do**

Example:

- Admin → Full access
- User → Limited access

1.6 Non-Repudiation

- Ensures that **sender cannot deny sending the message**

Technique:

- Digital Signature

Example:

Online contracts cannot be denied later

1.7 Accountability

- Tracks user activities

Example:

Logs showing who accessed which file

THREATS AND VULNERABILITIES

Threat

A **potential cause of harm** to a system

Types of Threats

(A) Passive Attacks

- No modification of data
- Difficult to detect

Types:

1. **Eavesdropping**
 - Listening to communication
2. **Traffic Analysis**
 - Observing communication patterns

(B) Active Attacks

- Modify or disrupt data

Types:

1. Masquerade Attack

- Attacker pretends to be authorized user

2. Replay Attack

- Captured message is retransmitted

3. Message Modification

- Data is altered

4. Denial of Service (DoS)

- System is overloaded

2.3 Vulnerability

- Weakness in system design or implementation

Examples:

- Weak passwords
- Open ports
- Outdated software
- Poor encryption

2.4 Relationship

Threat + Vulnerability → Attack

Example:

- Threat: Hacker
- Vulnerability: Weak password
- Attack: Account hacked

BASIC SECURITY MECHANISMS

These mechanisms are used to achieve security goals:

1. Encryption

(A) Symmetric Key Encryption

- Same key for encryption and decryption
- Fast but less secure

Example:

Sender and receiver share one secret key

(B) Asymmetric Key Encryption

- Uses two keys:
 - Public Key
 - Private Key
- More secure but slower

2. Hash Functions

- Converts input → fixed-size output
- One-way function

Properties:

- Cannot reverse
- Same input → same output
- Small change → large difference

3. Digital Signatures

- Created using sender's private key
- Verified using public key

Provides:

- Authentication
- Integrity

- Non-repudiation

4. Authentication Mechanisms

- Password-based
- OTP-based
- Biometric (fingerprint, face)

5. Access Control

Types:

1. DAC (Discretionary Access Control)
2. MAC (Mandatory Access Control)
3. RBAC (Role-Based Access Control)

6. Firewalls

- Filters network traffic

Types:

- Packet filtering firewall
- Stateful firewall
- Application-level firewall

7. Intrusion Detection System (IDS)

Types:

- Network-based IDS (NIDS)
- Host-based IDS (HIDS)

Function:

- Detect suspicious activity

8. Intrusion Prevention System (IPS)

- Detects and **prevents** attacks

9. Antivirus and Anti-malware

- Detect and remove malicious software

NETWORK SECURITY

Protecting **data, devices, and communication channels**

Key Components

(A) Cryptography

- Core part of security

Includes:

- Encryption
- Hashing
- Digital signatures

(B) Security Protocols

1. **SSL/TLS**
 - Secure communication over internet
2. **HTTPS**
 - HTTP + SSL/TLS
3. **IPsec**
 - Secure IP packets
4. **SSH**
 - Secure remote login

(C) Network Security Devices

- Firewalls
- Routers
- IDS/IPS

Security at Different Layers

Layer	Security Mechanism
Application	HTTPS, Email Security
Transport	TLS
Network	IPsec

Common Network Attacks

- 1. Man-in-the-Middle (MITM):** Attacker intercepts communication
- 2. Phishing:** Fake emails/websites to steal data
- 3. Malware:** Virus, worm, trojan
- 4. DoS/DDoS:** Overloads server

Security Services

- Confidentiality
- Integrity
- Authentication
- Access Control
- Non-repudiation